

Builder Basics — The Sloped Roof (Extension · Beginner)

Topic: Variables · **Course:** Builder Basics · **Type:** Extension Activity · **Level:** Beginner · **Time:** about 30 minutes

🧩 This is a bonus challenge, not a weekly lesson. Try it once you are happy building with loops.

1 · Predict 🧠

Open the **Sloped Roof** world. Do not run it yet.

It builds a roof from stairs, one square **ring** at a time. A ring is a square outline. The program stacks 4 rings, each on top of the last.

Each ring is **shorter** than the one below. A number called `roof_len` sets the side length. It starts at 7 and loses 2 each ring. So the rings shrink and rise to a peak.

What shape does this make? Circle one: flat square · stepped pyramid roof · single line

Why is each ring smaller than the one below? Circle one: to step up to a peak · to get wider · to dig down

2 · Watch the Variable Change

A **variable** is a name that holds a number. The number can **change** while the program runs.

Here it is `roof_len` — the side length of each ring. It starts at 7 and loses 2 each ring.

Fill in the table. The first one is done for you.

Layer	<code>roof_len</code> =
Layer 1 (bottom)	7
Layer 2	___
Layer 3	___
Layer 4 (top)	___

The widest ring is at the (circle one): bottom · top

3 · Think It Through

If `roof_len` stayed 7, what would the roof look like? Circle one: a flat box, all rings the same size
· a pointed peak · a deep hole

Which number makes the roof start wider at the bottom? Circle one: the starting 7 · the peak
block · the world name

Now run it for real.

1. Open the **Sloped Roof** world.
2. Type **slope** in the Minecraft chat and press enter to run the program.
3. Watch the rings build up, getting smaller each time, until they reach the peak.

Then change the starting number. Make `roof_len` start at **9** instead of 7, run it again, and see how the roof changes.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: ____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.